



BUYLOG SECTION 11

Panelboards

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ReliaGear lighting panelboards

Applications

Standards

All ABB lighting panelboards meet the latest revision of the following standards.

- National Electrical Code—Ref. Article 384
- UL67 panelboards: UL50 cabinets and boxes
UL943 GFCI
- UL489 molded case circuit breakers
- cULus listing for ReliaGear lighting panelboards
- cULus listing for ReliaGear non-service entrance panelboards
- International Building Code Seismic Certification
- California Building Code Seismic Certification
- NEMA PB1
- Federal Specifications
 - Panelboards, W-P-115c
 - Type 1—Circuit breaker equipped
 - Class 1—Panelboards
 - Class 2—Load centers
 - Molded case circuit breakers, WC-375B/GEN

Application

The following classifications and limitations of panelboards have been established by the Underwriters Laboratories and the National Electrical Code. Note—“an overcurrent protective device is a circuit breaker pole or single fuse”. Panelboards have no fire wall ratings. All 50/60 Hz rated. There is no limitation as to the number and rating of branch circuits, except as determined by available enclosures.

Interrupting ratings—circuit breakers

Panelboards have integrated short circuit ratings. When fully rated, the rating is that of the lowest rated device in the panelboard. When series connected rated, the rating is that of the main device in panel (or remote line side protected device) and branch-tested/UL Listed combination.

Short-circuit ratings—fusible switch units

The interrupting rating of the fuse must equal or exceed the short-circuit rating of the switch. If it is lower, then the interrupting rating of the switch is the same as the fuse. Switches have no short-circuit rating if renewable fuses are used.

Seismic ratings

All ReliaGear Lighting Panelboards have been tested and certified to meet the seismic requirements of 2018 International Building Code (IBC) as well as the 2019 California Building Code (CBC)

Selective coordination

NFPA 70, the National Electrical Code (NEC), requires overcurrent devices to be selectively coordinated when applied in emergency standby systems (Article 700), legally required standby systems (Article 701), Critical Power Systems (Article 708) and when supplying multiple elevator circuits (620.62). The NEC defines the performance standard of selective coordination in Article 100, Definitions. Beginning with the definition in effect with the 2014 NEC, the



combinations of circuit breakers that can comply with this standard are limited. Those limitations include the number of circuit breaker poles, current ratings of either the line side or load side circuit breaker, and the maximum interrupting current that selective operation extends to. These limitations can affect the selection of circuit breakers used in a panelboard. ABB has documented selective pairs of their molded case circuit breakers in publication 1SDC210066D0201. This publication should be consulted when applying panelboards in the applications noted above.

Features

- Symmetrical design, no required top or bottom mountings
- Wide, easy-to-install galvanized enclosures with removable endwalls
- Flush or surface mounting for NEMA Type 1 enclosures
- Standard concealed mounting hardware and hinges
- Interiors that allow “straight-in” wiring
- Split neutral for 400A and higher panelboards
- Branch-bus direct connection
- Captive hardware on branch circuit breakers
- Short circuit ratings allow up to 100KA @ 480Y/277Vac; 200KA @ 240Vac (depending on type of panelboard)
- Main bus ratings of 125 to 800 amps copper, 125 to 600 amps aluminum
- Tmax XT vertically mounted main circuit breakers with fixed thermal magnetic and adjustable trip units available
- Bus-connected SPD for maximum surge protection
- Optional door-in-door or front-hinged-to-box door
- Enclosures available in NEMA Type 1, Type 3R/12, Type 4/4X painted galvanneal or 316 grade stainless steel
- Optional metering features
- Front options: stainless steel front
- Build America, Buy America, 2 CFR Part 184 options available
- Latch/lock options: best lock, national lock/corbin 60 key, corbin latch bolt 15767, yale lock, replacement locks

ReliaGear lighting panelboards

Types

Type RL

Service information:

1P, 3W-120/240 Vac
 3P, 3W-240 Vac
 3P, 4W-240/120 Vac, 208/120 Vac
 Fully rated: 65kAIC at 240V
 Series rated: 200kAIC at 240V

Main circuit breakers:

100A-THQB, THHQB, TEY
 125A-XT1
 225A-A2
 250A-XT4
 400A-XT5
 600A-XT5
 800A-XT6

Main lug:

125-800A

Branch circuit breakers (plug-in):

Amperage-15-100A
 Poles-1, 2, 3
 Types-THQL, THHQL, TXQL

Subfeeds:

Amperage-15-600A
 Poles-2, 3
 Types- A2, XT1, XT4, XT5

Enclosures

Height-25.5, 31.5, 37.5, 43.5, 49.5, 55.5, 64.5, 76.5, 82.5, 88.5
 Width-20"
 Depth-5.81"



Type RQ



Type RL

Type RQ

Service information:

1P, 3W-120/240 Vac
 3P, 3W-240 Vac
 3P, 4W-240/120 Vac, 208/120 Vac
 Fully rated: 65kAIC at 240V
 Series rated: 200kAIC at 240V

Main circuit breakers:

100A-THQB, THHQB, TEY
 125A-XT1
 225A-A2
 250A-XT4
 400A-XT5
 600A-XT5
 800A-XT6

Main lug:

125-800A

Branch circuit breakers (bolt-on):

Amperage-15-100A
 Poles-1, 2, 3
 Types-THQB, THHQB, TXQB

Subfeeds:

Amperage-15-600A
 Poles-2, 3
 Types- A2, XT1, XT4, XT5

Enclosures

Height-25.5, 31.5, 37.5, 43.5, 49.5, 55.5, 64.5, 76.5, 82.5, 88.5
 Width-20"
 Depth-5.81"

ReliaGear lighting panelboards

Types

Type RE

Service information:

1P, 3W-120/240 Vac
 3P, 3W-240 Vac
 3P, 4W-480/277 Vac, 208/120 Vac, 240/120 Vac
 Fully rated: 18kAIC at 480Y/277V, 65kAIC at 240V
 Series rated: Reference panel configuration in empower
 or DET-008

Main circuit breakers:

100A-TEY, TEYF
 125A-XT1
 250A-XT4
 400A-XT5
 600A-XT5
 800A-XT6

Main lug:

125-800A

Branch circuit breakers (bolt-on):

Amperage-15-100A
 Poles-1, 2, 3
 Types-TEY, TEYF

Subfeeds:

Amperage: 15-600A
 Poles: 3*
 Types: XT1, XT4, XT5

*3 poles can be used as 2 poles

Enclosures

Height-25.5, 31.5, 37.5, 43.5, 49.5, 55.5, 64.5, 76.5, 82.5, 88.5
 Width-20"
 Depth-5.81"

Type RS

Service information:

1P, 3W-120/240 Vac
 2P, 2W-125 Vdc, 250 Vdc
 3P, 3W-240 Vac
 3P, 4W-480/277 Vac, 125/250 Vdc, 208/120 Vac, 240/120 Vac
 Fully rated: 18kAIC at 125/250 Vdc, 65kAIC at 480Y/277V,
 100kAIC at 240V
 Series rated: Reference panel configuration in empower
 or DET-008

Main circuit breakers:

100A-TEYD, TEYH, TEYL
 125A-XT1
 250A-XT4
 400A-XT5
 600A-XT5
 800A-XT6

Main lug:

125-800A

Branch circuit breakers (bolt-on):

Amperage-15-125A
 Poles- 1, 2, 3
 Types- TEYD, TEYH, TEYL

Subfeeds:

Amperage: 15-600A
 Poles: 3*
 Types: XT1, XT4, XT5

*3 poles can be used as 2 poles

Enclosures

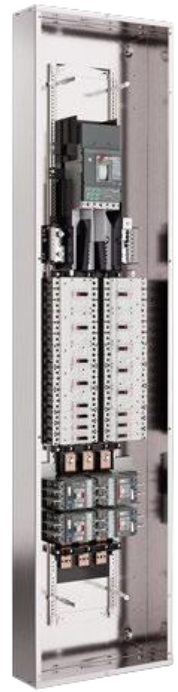
Height-25.5, 31.5, 37.5, 43.5, 49.5, 55.5, 64.5, 76.5, 82.5, 88.5
 Width-20"
 Depth-5.81"



Type RE



Type RS



Type RD

Type RD

Service information:

2P, 125 Vdc, 250 Vdc
 3P, 3W-480 Vac, 600 Vac
 3P, 4W-208/120 Vac, 125/250 Vdc, 480/277 Vac, 600/347 Vac
 Fully rated: 35kAIC at 125/250 Vdc, 42kAIC at 240V,
 42kAIC at 600Y/347V, 42kAIC at 480V
 Series rated: Reference panel configuration in empower
 or DET-008

Main circuit breakers:

150A-XT4
 225A-XT4
 400A-XT5
 600A-XT5

Main lug:

125-600A

Branch circuit breakers (bolt-on):

Amperage-15-100A (FB 1P, 2P, 3P)
 200A Max. double branch
 Types: FB

Subfeeds:

Amperage: 15-400A
 Poles: 3
 Types: XT2 (TMA, DIP, Hi-Touch), FB, XT4 (TMF, DIP, Hi-Touch),
 XT5 (DIP, Hi-Touch)

Enclosures

Height-31.5, 37.5, 43.5, 49.5, 55.5, 64.5, 76.5, 82.5, 88.5
 Width-20"
 Depth-5.81"

ReliaGear lighting panelboards

Types

Panelboard types

Item	ReliaGear				
	RL Page 11-6	RQ Page 11-6	RE Page 11-7	RS Page 11-7	RD Page 11-7
Max. voltage	240Vac	240Vac	480Y/277Vac 125/250Vdc	480Y/277Vac 125/250Vdc	600Vac
Max. main lug amperes	800A	800A	800A	800A	600A
Max. main circuit breaker or switch amperes	800A	800A	800A	800A	600A
Main devices	THQB, THHQB, TEY	THQB, THHQB, A2,TEY	TEY, TEYF	TEYD, TEYH, TEYL	XT4, XT5
	A2, XT1, XT4, XT5, XT6		XT1, XT4, XT5, XT6		-
Branches max. amps	100A	100A	100A	100A	100A-1 ph, 2ph
Branch devices	THQL, THHQL	THQB, THHQB, TXQB (Bolt-on)	TEY, TEYF	TEYD, TEYH, TEYL	FB
Subfeed circuit breaker types	A2, XT1, XT4, XT5	A2, XT1, XT4, XT5	XT1, XT4, XT5	XT1, XT4, XT5	XT2, XT4, XT5

ReliaGear standard main circuit breaker types and ratings

IC ratings	Voltage	Panel type															
		RQ/RL					RE/RS							RD			
		100A	225A	400A	600A	800A	100A	125A	225A	400A	600A	800A	150A	225A	400A	600A	
18	125/250 Vdc	-	-	-	-	-	-	-	-	-	-	-	XT4N	XT4N	XT5N	XT5N	
35	125/250 Vdc	-	-	-	-	-	-	-	-	-	-	-	XT4N	XT4N	XT5N	XT5N	
10	240	THQB	A2A	-	-	XT6N	-	-	-	-	-	-	-	-	-	-	
22	240	THHQB	A2N	-	XT5N	-	-	-	-	-	-	-	-	-	-	-	
65	240	TEY/ XT1S	XT4N	XT5N	XT5N	-	-	-	XT4N	XT5N	XT5N	-	-	-	-	-	
100	240	XT1H	XT4S	XT5S	XT5S	-	-	-	-	-	-	-	-	-	-	-	
200	240	-	XT4L	XT5L	-	-	-	-	-	-	-	-	-	-	-	-	
14	480Y/277	-	-	-	-	-	TEY	-	-	XT5N	XT5N	XT6N	XT4N	XT4N	XT5N	XT5N	
35	480Y/277	-	-	-	-	-	-	-	-	-	-	-	XT4S	XT4S	XT5N	XT5N	
18	480Y/277	-	-	-	-	-	-	-	-	-	-	-	XT4N	XT4N	XT5N	XT5N	
25	480Y/277	-	-	-	-	-	-	XT1N/ TEYD	XT4N	-	-	-	XT4N	XT4N	XT5N	XT5N	
65	480Y/277	-	-	-	-	-	-	XT1H/ TEYL	XT4H	XT5H	XT5H	-	-	-	-	-	
42	480Y/277	-	-	-	-	-	-	-	-	-	-	-	XT4H	XT4H	XT5S	XT5S	
100	480Y/277	-	-	-	-	-	-	XT1L	XT4L	-	-	-	-	-	-	-	
14	480	-	-	-	-	-	-	-	-	-	-	-	XT4N	XT4N	XT5N	XT5N	
42	480	-	-	-	-	-	-	-	-	-	-	-	XT4H	XT4H	XT5S	XT5S	
18	600	-	-	-	-	-	-	-	-	-	-	-	XT4N	XT4N	XT5N	XT5N	
25	600	-	-	-	-	-	-	-	-	-	-	-	XT4S	XT4S	XT5S	XT5S	
42	600Y/347	-	-	-	-	-	TEYL	TEYL	XT4N	XT5N	XT5N	-	XT4L	XT4L	XT5L	XT5L	

ReliaGear lighting panelboards

Terminal lugs

Molded case circuit breakers

Frame	Poles	Lug kit number ¹	Cable(s) per lug	Cable Range
XT6	3	KXT6CUAL3X400K-3PC	3	Cu Al 3x2/0AWG-400kcmil
XT6 (750MCM)	3	KXT6CUAL2X750KCC-3	2	500kcmil - 750 kcmil ³
XT5	3	KXT5CUAL2X500K-3PC	2	Cu Al 2x2/0AWG-500kcmil
XT5 (750MCM)	3	KXT5CUAL2X750KC-3	2	500kcmil - 750 kcmil ³
XT4-250A	3	KXT4CUAL3-3PC	1	Cu Al 1x3/0 AWG-350 kcmil ²
XT4 (<250A)	3	KXT4CUAL2-3PC	1	Cu Al 1x4 AWG-300 kcmil
XT1	3	KXT1CUAL1-3PC	1	Cu Al 1x14-2/0 AWG
A2	3	1SDA069983R1 (3pole - 3pcs lug)	1	Cu 1x1 AWG-250kcmil
				Al 1x2/0 AWG-300
A2	2	1SDA069982R1 (2pole - 2pcs lug)	1	Cu 1x1 AWG-250kcmil
				Al 1x2/0 AWG-300

¹ Kit contains 3pcs lug

² External solution: lugs to be mounted on EF terminals supplied in the kit

³ The lug kit will come with 2 sets of cable set screws. One is for 600 MCM and smaller cable and the other is for cable greater than 600 MCM. Follow the instructions that are included with the kit. If you are upgrading the existing lugs to the 750 MCM lugs, the customer, AHJ (authority having jurisdiction) and/or inspector will need to make sure the panel is compliant with NEC and UL cable bending space. ABB is not responsible for the addition of these lugs in existing panels.

ReliaGear lighting panelboards

Enclosures



NEMA 1

Front options sold separately



NEMA 3R

NEMA 4, 4X & 12
painted galvanealNEMA 4, 4X & 12
316 stainless steel

Enclosure

Box Height	NEMA 1	NEMA 1	NEMA 3R		NEMA 4, 4X & 12 Painted Galvaneal	NEMA 4, 4X & 12 Painted Galvaneal	NEMA 4, 4X & 12 Stainless Steel	
	20" Wide	30" Wide	20" Wide	30" Wide	20" Wide	30" Wide	20" Wide	30" Wide
25.5"	–	–	–	–	–	–	AB254S	AB254DWS
31.5"	AB31B	AB31BW	AB313	AB313DW	AB314	AB314DW	AB314S	AB314DWS
37.5"	AB37B	AB37BW	AB373	AB373DW	AB374	AB374DW	AB374S	AB374DWS
43.5"	AB43B	AB43BW	AB433	AB433DW	AB434	AB434DW	AB434S	AB434DWS
49.5"	AB49B	AB49BW	AB493	AB493DW	AB494	AB494DW	AB494S	AB494DWS
55.5"	AB55B	AB55BW	AB553	AB553DW	AB554	AB554DW	AB554S	AB554DWS
64.5"	AB64B	AB64BW	AB643	AB643DW	AB644	AB644DW	AB644S	AB644DWS
76.5"	AB76B	AB76BW	AB763	AB763DW	AB764	AB764DW	AB764S	AB764DWS
82.5"	AB82B	AB82BW	AB823	AB823DW	AB824	AB824DW	AB824S	AB824DWS
88.5"	AB88B	AB88BW	AB883	AB883DW	AB884	AB884DW	AB884S	AB884DWS



Standard



Standard with quarter turn lock



Door within door



Front hinged to Box

Enclosure front

Front Height	Standard (20" wide)		Door Within Door (20" wide)		Front Hinged To Box (20" wide)		Standard (30" Wide)	
	Flush	Surface	Flush	Surface	Flush	Surface	Flush	Surface
31.5"	AF31F	AF31S	AF31FP	AF31SP	AF31FD	AF31SD	AF31FW	AF31SW
37.5"	AF37F	AF37S	AF37FP	AF37SP	AF37FD	AF37SD	AF37FW	AF37SW
43.5"	AF43F	AF43S	AF43FP	AF43SP	AF43FD	AF43SD	AF43FW	AF43SW
49.5"	AF49F	AF49S	AF49FP	AF49SP	AF49FD	AF49SD	AF49FW	AF49SW
55.5"	AF55F	AF55S	AF55FP	AF55SP	AF55FD	AF55SD	AF55FW	AF55SW
64.5"	AF64F(T)	AF64S(T)	AF64FP(T)	AF64SP(T)	AF64FD(T)	AF64SD(T)	AF64FW(T)	AF64SW(T)
76.5"	AF76F(T)	AF76S(T)	AF76FP(T)	AF76SP(T)	AF76FD(T)	AF76SD(T)	AF76FW(T)	AF76SW(T)
82.5"	AF82F(T)	AF82S(T)	AF82FP(T)	AF82SP(T)	AF82FD(T)	AF82SD(T)	AF82FW(T)	AF82SW(T)
88.5"	AF88F(T)	AF88S(T)	AF88FP(T)	AF88SP(T)	AF88FD(T)	AF88SD(T)	AF88FW(T)	AF88SW(T)

(T) Fronts with quarter turn lock for applications with XT5 and/or XT6 circuit breakers. Applicable for NEMA 1 enclosures only.

ReliaGear lighting panelboards

Pricing and ordering through empower, distributors or sales

Information required to price and order a panelboard

- Short-circuit rating (10kA, 18kA, etc.)
- Service entrance label (Yes) or (No)
- Service (3-ph, 4-w 208Y/120; 3-ph, 3-w 480 volts, etc.)
- Entrance of incoming line (top) or (bottom).
(Bottom supplied as standard)
- Trim (surface) or (flush)
- Incoming wire size (500kcmil, 250kcmil, etc.)
- Incoming number of wires per phase (1, 2, 3, etc.)
- Wire material (copper or aluminum)
- Main type (main lugs only, circuit breaker, fusible switch, etc.)
- Amperage of main bus
- Frame of main circuit breaker (XT5, XT6, etc.)
(if applicable)
- Options to mains (shunt trip, lighting contactor, etc.)
- Equipment ground (optional)
- Branches
 - Amp rating (20, 30, 50, etc.)
 - Poles (1, 2, or 3)
 - Frame (THQB, TEY, etc.)
 - Quantity (1, 10, 15, etc.)
- Options:
 - Interior (copper bus, 200% rated neutral, etc.)
 - Box (painted, increased gutter, etc.)
 - Front (door in door, etc.)
 - Ground fault protection (yes) or (no)
- Type of panel (RQ, RE, etc.)

Pricing and layout for factory assembled and unassembled panelboards through empower.

<https://empower.abb.com>

How to select a ReliaGear® Merchandised, unassembled panelboard

Total the following components:

Step 1: Select Interior

Step 2: Select Box

Step 3: Select Front

Step 4: Select main and/or sub-feed breaker kit(s)

Step 5: Select main and/or sub-feed breaker(s)

Step 6: Select main and/or feed-thru lug kit(s) and accessories

Step 7: Select accessories

Step 8: Select bolt-on branch circuit breakers

Please consult your local distributor for net pricing and current stock levels.

For additional details on selecting ReliaGear Merchandised panelboards, please refer to publication 1TQC173700E0051.

ReliaGear lighting panelboards

Product options

Enclosure options

Box extensions—For additional end gutter space or conduit skirt applications—see page 11-14.

Equipment grounds—factory supplied with panelboard

Description	Product Number
Field installed kits	
Standard bonded to box—for each 12 branch positions	TGL2
Copper bonded to box—for each 12 branch positions	TGC2
Standard—isolated/insulated—for each 12 branch positions	EGS12
Copper—isolated/insulated—for each 12 branch positions	EGC12
Main lug for above terminal kits	TGL20

Optional equipment grounds

Description	Product Number
Aluminum Extruded Bonded	AEBG
Copper Extruded Bonded	AEBGC
Aluminum Extruded Isolated	AEIG
Copper Extruded Isolated	AEIGC
Copper Isolated/Bonded	ASPGIBC

THQB/THHQB/THQL/THHQL/TEY filler plates

Product Number
TQLFP1

Filler plates

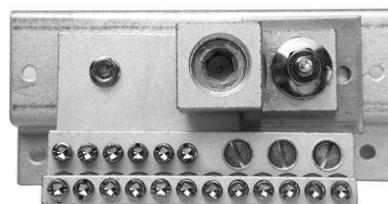
Description	Product Number
T(HE)D/SE/FB (for legacy A series panels)	TEDFP1
XT2/FB (for ReliaGear panels) (contains 10 filler plates)	XT2FBFP10

Circuit breaker mounting¹ hardware kits

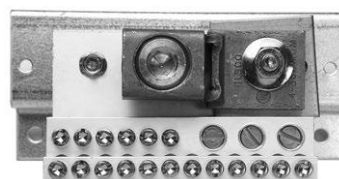
Description	Product Number
Circuit breaker type TED/THED4/SE	ASPTED3P
Circuit breaker type FB	ASPFBRD3P
Circuit breaker type Formula A2	ASPA23P
Circuit breaker type Tmax XT1— for mounting 3 poles	ASPXT13P ²
Circuit breaker type Tmax XT2— for mounting 3 poles	ASPXT23P ²
Circuit breaker type Tmax XT4— for mounting 3 poles	ASPXT43P ²

¹ Use to mount circuit breaker in existing space.

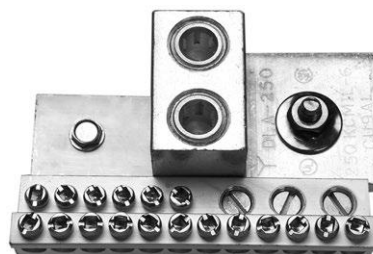
² Includes screws, washers and ReliaGear panelboard connectors.



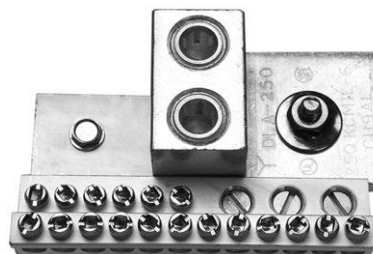
AEBG



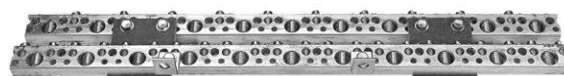
AEBGC



AEIG



AEIGC



ASPGIBC

ReliaGear lighting panelboards

Accessories

Field installed kits/replacement parts
Order by product number from factory

Endwall kits

Field installed. For standard 20"w x 5.81"d boxes.

Product Number	Description	Qty.
ABEW2	Blank	1
AKEW2	Knockout	1
ABEW3	Blank - 30" x 5 3/4"	1
AKEW3	Knockout - 30" W	1

Panelboard locks

Description	Product Number
T-Handle Quarter Turn Kit ¹	ASPQTRT
Flush Quarter Turn Kit ¹	ASPQTRK
Replacement Lock with Std. Key	569B737P1
Additional Keys for Above Lock	569B737P5
Yale Lock Kit	ASPYALE47
Corbin Lock Kit	ASPCORBNTU1
Replacement Lock with ABB75 Key	569B737P2

¹ The quarter turn kits are only compatible with factory-built quarter turn fronts. They cannot be used to convert other locks (standard, Corbin, etc.) to quarter turn. Keys are interchangeable with standard, Corbin, etc.

Locking devices

Frame Style	Description	Product Number
Padlocking Device (single padlock)		
Q	THQB, THHQB, THQL, THHQL	THP100
Q	GFT2, GFEP, AF2, DF2	THP100E
E	TEY	TEYPLD1
Formula	Formula A2 3 pole	KA2LD
Formula	Formua A2 2 pole	KA2LDOR
Tmax XT	XT1	KXTBPLOPCL
Tmax XT	XT2 and XT4	KXTCPLLOPCL
Tmax XT	XT5	KXT5PLOPLC
Tmax XT	XT6	KXT6PLOPLC

Handle Locking (nonpadlocking)

Q	THQB, THHQB, THL, THHQL	THL103
E	TEY	TEYLD1
	Filler plate for Q and TEY breakers	TQLFP1
Q/E	Safety catch for trough covers	ACHK
	Gasketing Kit	AGSK

Main breaker service entrance lug cover kit

Breaker Type	Poles	Product Number
Formula A2	2	A2P2SB1
Formula A2	3	A2P3SB1
Tmax XT1	3	XT1P3SB1
Tmax XT4	3	XT4P3SB1
Tmax XT5	3	XT5P3SB1



T-Handle Quarter Turn Kit



Replacement Lock with Standard Key - Black



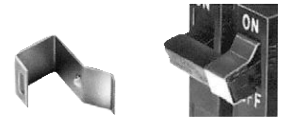
Corbin Lock



Replacement Lock with ABB75 Key - Red



Fixed padlock in the open position - PLL



Handle locking THL103

ReliaGear lighting panelboards

Accessories

Field installed kits/replacement parts
Order by product number from factory

Panelboards parts

Description	Product Number
Directory Card	139C5612P3
Circuit Numbering Strips – 1–48	569B806G1
49–84	569B806G2
85–126	569B806G3
Adhesive Backed Lamicaid Nameplate 3/4 in. x 3 in.	315A7190P1
Metal Directory Card Holder	139C5491G1
Directory Card Holder	139C5491P4
Delta Hi-leg Conversion Kit, to Add B-Phase Plug on RL Panels	APHBL
Delta Hi-leg Conversion Kit, to Add B-Phase Bolt on RE/RQ Panels	APHBQ
NEMA 3R/12 Tamper Proof Tork Screw Kit	NEMATRX
RD 25 to 65 kAIC Barrier kit	ASP25AD65KA ¹
Service Entrance Kit	ASPSERENT
RQ/RL/RE Rail Bracket	ASPAQLEBKT
Front Flush Adjust Kit	ASPFLUSHADJ
RE Front Mounting Kit	139C5720G3
RQ/RL Front Mounting Kit	139C5720G6
RD Front Mounting Kit	139C5720G9
Front Hinge to Box Mounting Kit	139C5700G6
Front Extension Mounting Kit	139C5700G11

¹ Included in factory assembled panels—AD panels with Spectra branch circuit breakers.

Permanent circuit number kits

Product Number		Description
RQ, RL, RE	AD	
APN48	APN48AD	No's 1–48
APN84	APN84AD	No's 43–84
APN126	APN126AD	No's 85–126

ACARE Kit²

Description	Product Number
ACARE spare parts kit for ReliaGear panelboard installation and maintenance	ACARE

² ACARE kit includes: (50) 1/4–20 philp HD THD form screw, (50) directory cards, (50) 1/4–20X1/2 phillips, (50) #1–42 circuit marker strip, (50) #43–84 circuit marker strip, (50) service disconnect labels, (25) 1/4–20 3/8 SLT. HD., (50) #10–32 X 7/16" slot hex wsh, (15) key # B–135A for black locks, (25) rating label books, (50) 1/4–20X1 1/2 phil thd roll, (50) 1/4–20 speed nut.

Box extensions

Bolts to ReliaGear box with or without box endwall in place. Extensions can be combined to obtain lengths greater than 18 and 24 inches.

Box Width and Depth	Box Mounting	Box Extension Length (Inches)	Box Extension Product Number
20 x 5.81	Flush	9	ABX2509F
		18	ABX2518F
		24	ABX2524F
	Surface	9	ABX2509S
		18	ABX2518S
		24	ABX2524S
		31	ABX2531S
		37	ABX2537S
		43	ABX2543S
		49	ABX2549S
		55	ABX2555S
		64	ABX2564S
		76	ABX2576S
30 x 5.81	Flush	18	ABX3518F
		24	ABX3524F
	Surface	18	ABX3518S
		24	ABX3524S
30 x 7.81	Flush	18	ABX3718F
		24	ABX3724F
	Surface	18	ABX3718S
		24	ABX3724S

Box Extensions Covers Only

10 covers per kit.

Description	Product Number
9" Covers Surface	ASPABX09S
9" Covers Flush	ASPABX09F
18" Covers Surface	ASPABX18S
18" Covers Flush	ASPABX18F
64" to 76" Covers Surface	ASPABX20S
64" to 76" Covers Flush	ASPABX20F

ReliaGear lighting panelboards

Branch circuit monitoring

The Branch Circuit Monitoring (BCM) unit offers a cost-effective, integrated solution for power monitoring and submetering in ReliaGear Lighting Panelboards. Featuring ANSI C12.20 0.5% revenue-grade metering accuracy, the BCM meter is suitable for tenant billing and cost allocation. With outstanding accuracy, the BCM unit monitors key electrical parameters of the various branch circuits within the panelboard. This data can be transmitted via multiple communication protocols to analyze usage, identify potential cost-saving measures, and improve load management. The BCM includes a local display with captive touch screen technology to provide faster startups and ease of access to critical information.

Features

- Solutions up to 800A
- Revenue grade, ANSI C12.20 0.5% accuracy
- Encapsulated Solid Core CT's that cover branch breakers from 1/2A –120A
- Ships fully assembled from the factory in the enclosure
- Solid Core monitors up to 84 circuits
- Configurable alarm thresholds improve load management
- Ability to set the orientation and numbering of the circuits
- 1-, 2-, 3-pole circuit breaker support
- Local display captive touch screen technology
- Front access via USB Micro B port
- Modbus RTU via 2-wire RS485 communications
- Modbus TCP, Bacnet IP (Ethernet RJ45 10/100 Mbs)

General measurements

- +/- kWh
- +/- kvarh
- kVAh
- Load curves/demand profiles

Multi-measurement

- I1, I2, I3, In
- ΣP , ΣQ , ΣS , ΣPF
- P, Q, S, PF per phase
- Current unbalance
- Phase angle
- Voltage measurement – phase to phase
- Ph/N unbalance, Ph/Ph unbalance
- Multi-load current measurement
- Operation across 4 quadrants
- Predictive power



Power quality

- THDi1, THDi2, THDi3, THDiN, THDisys and harmonics up to 63rd for voltage and current
- K-factor
- Crest factor I1, I2, I3
- Overcurrent

Alarms

- Measurement thresholds
- System alarms
- Boolean combination of alarms

Trends

- Average values

Advanced features

- Auto correct technology

References

[ISQC173002B0201](#) for additional information
empower for configuration/quotation purposes

ReliaGear lighting panelboards

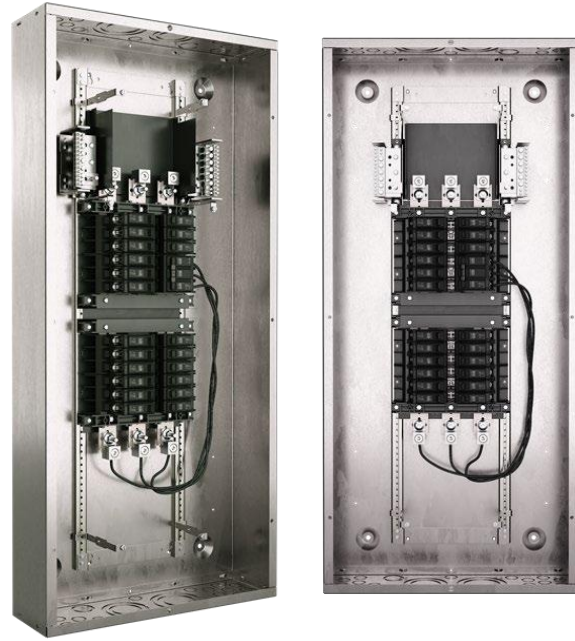
Title 24 solutions

California's Title 24, Part 6, Building Energy Efficiency Standards, §130.5(b) requires electrical systems are to be arranged to allow metering of electrical loads by load type or other classifications. While the meters themselves do not need to be installed, electrical equipment that is an intended location for this metering must be able to allow future current and voltage sensing. Section 130.5(b) applies to new electrical system installations or when complete electrical systems are replaced. The requirement does not apply to modifications made in existing electrical systems, such as adding a new circuit breaker to an electrical panel

Branch Circuit Monitoring (BCM) upgradable panels are a practical, cost effective way to meet Title 24 Part 6 §130.5(b) requirements. This method provides freedom to place branch devices anywhere in a panel regardless of load type. This means that panels can be installed with less labor since there is no further time dedicated to validating the construction of a complex layout.

Branch Circuit Monitoring (BCM) upgradable panelboards allow the ability to meter each branch circuit individually without the complexity and cost of physically grouping similar branch devices together within the panel. Title 24 compliance can be attained with BCM upgradable panels no matter where a branch device is located in the panel, provided a single load type is wired per branch. The BCM upgradable panelboards provide more installation flexibility and better density per panel than disaggregated load monitoring. The field upgrade kit can be installed after the panel is commissioned. The kit includes a meter and split core CTs. BCM upgradable panels retain density and flexibility when adding additional branch devices regardless if metering has been implemented.

Split bus panels are another cost-effective solution for disaggregating multiple type of loads that can be metered separately in the future using additive/subtractive current transformer wiring techniques. Up to seven sections can be configured in the panel, with space in between each section for future CTs. The quantity of the branches in each section is flexible and can be any multiple of six 1-pole branches.



Split Bus

Split bus configurations

- Incoming type:
 - Lighting contactor, main lug or main breaker
- Panelboard selections
 - RQ/RL up to 225A main breaker or main lug, max 22 KAIC at 240V
 - RE up to 125A main breaker or 225A main lug, max 18 KAIC at 480V
- Main breakers:
 - Formula A2, XT1, THQB, TEY, TEY(D/H/L)
 - Feeders: THQB, TEY
- Sub-feeds:
 - Formula A2, THQB, TEY, TEY(D/H/L) TED, XT1 Max rating of 22 KAIC at 240V and 208/120V and 18 KAIC @480V.
- 9 configurations available in empower

Features

- Solutions up to 800A
- Offers 3 solutions: BCM upgradable panels, field upgrade kits or split bus panels
- 1, 2, 3 pole circuit breaker support

ReliaGear merchandised, unassembled lighting panels

Pricing and ordering through empower, distributors or sales

100-600A (600A main circuit breaker not available)
240 Vac 1 or 3 phase or 480y/277 vac 3 phase
Order by product number from the customer service center

Step 1: Select interior

Select the interior by voltage, number of circuits and ampacity.
 Identify the box/front height for interior selected in steps 2 and 3.



Feed-thru copper bus

Voltage	Rating (amps)	No. of circuits	Product number ¹	Box/front height (in.)	No. of TGL2 ground bars required ²
240 V AC, 1-Phase	100-225	30	AQU1302RCXAXT1 B4	43.5	3
		42	AQU1422RCXAXT1 B4	49.5	4
	400	42	AQU1424RCXAXT1 B4	76.5	4
		42	AQU1426RCXAXT1 B4	76.5	4
208Y/120 V AC, 3-Phase	100-225	30	AQU3302RCXAXT1 B4	43.5	3
		42	AQU3422RCXAXT1 B4	49.5	4
	400	42	AQU3424RCXAXT1 B4	76.5	4
		42	AQU3426RCXAXT1 B4	76.5	4
480Y/277 V AC, 3-Phase	100-225	30	AEU3302RCXAXT1 B4	43.5	3
		42	AEU3422RCXAXT1 B4	49.5	4
	400	42	AEU3424RCXAXT1 B4	76.5	4
		42	AEU3426RCXAXT1 B4	76.5	4

¹ See table below for TGL20 ground lug quantities.

² For isolated ground, use EGS12. When using the EGS12, 5 and 7 ground lugs (TGL20s) are required for 30 and 42 circuit panels respectively.

³ 600 A available as main lug only.

TGL20 ground lugs required by panel type

Interior type	100-225 A	400 A	600 A
Main lug only	1	1	2
Main lug and sub-feed	2	2	4
Main lug and feed-thru	2	2	4
Main breaker only	1	1	-
Main breaker and sub-feed	2	2	-
Main breaker and feed-thru	2	2	-

ReliaGear merchandised, unassembled lighting panels

Pricing and ordering through empower, distributors or sales

Steps 2–3: Select box and front

Select a 20" wide box of correct height based on interior selected in Step 1.

- Boxes come with blank endwalls.



NEMA 1



NEMA 3R

Box height (in.)	NEMA 1 product number	NEMA 3R product number
43.5	AB43B	AB433
49.5	AB49B	AB493
76.5	AB76B	AB763

For NEMA 1 enclosures only, select a 20" wide front of correct height based on interior selected in Step 1 and box selected above.

- Standard fronts are equipped with concealed hinges and trim adjusting screws hinges, trim adjusting screws and quarter-turn locks.
- Door within door fronts are convenient for easy access to equipment from the front of the panel because they allow access to the gutters without removing the front.
- Front hinged to box fronts are similar to door within door fronts for convenient access to gutters, but four screws must be removed to access the outer door.



Standard



Door within door



Standard with quarter-turn lock

Front height (in.)	Standard	Standard	Door within door	Front hinged to box
	Flush product number	Surface product number	Surface product number	Surface product number
43.5	AF43F	AF43S	AF43SP	AF43SD
49.5	AF49F	AF49S	AF49SP	AF49SD
76.5 ¹	AF76FT	AF76ST	AF76SPT	AF76SDT

¹ 76.5" fronts include quarter-turn lock, enabling applications with XT5 circuit breakers.

NOTE: Spectra has been obsoleted but these kits are still available to support existing installations.